

HORNSBY GIRLS HIGH SCHOOL



2010 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using a black or blue pen
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown for every question
- Begin each question in a new booklet

Total marks (84)

- Attempt Questions 1 – 7
- All questions are of equal value

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Total Marks**Attempt Questions 1–7****All Questions are of equal value**

Begin each question in a new booklet, writing your student number and question number in the boxes indicated. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.	Marks
(a) The interval AB , where A is $(-3, 4)$ and B is $(1, -2)$, is divided externally in the ratio 1: 3 by the point $P(x, y)$. Find the values of x and y .	2
(b) Differentiate $\cos^{-1}(x^3)$ with respect to x .	2
(c) Prove that, if $x^4 - x^3 + kx - 4$ has a factor of $(x + 1)$, then it also has a factor of $(x - 2)$.	2
(d) Solve $\frac{x+4}{x-2} \geq 3$ for x .	3
(e) Use the substitution $u = 2x + 1$ to evaluate $\int_0^1 \frac{4x}{2x+1} dx$.	3

Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Solve (giving your answer to nearest degree):
 $3 \cos x + 5 \sin x = 5$ for $0^\circ \leq x \leq 360^\circ$ 3

- (b) The curves $y = x^2$ and $y = 2x$ meet at $x = 2$. Find the angle between these curves at this point of intersection. 2

- (c) Find the primitive function of $2 \sin^2 x$. 2

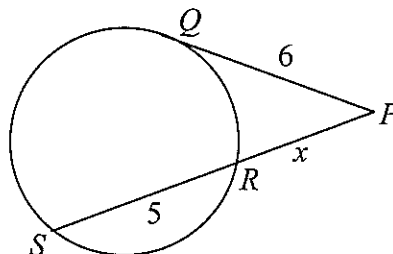
- (d) In a large school, 5% of the students have blond hair. A group of 10 students is randomly chosen.

What is the probability that:

- (i) exactly one student has blond hair? 1

- (ii) at least 2 students have blond hair? 2

(e)



- PQ is a tangent to a circle QRS , while PRS is a secant intersecting the circle at R and S , as shown in the diagram. 2

Given that $PQ = 6$, $RS = 5$ and $PR = x$, find the value of x .

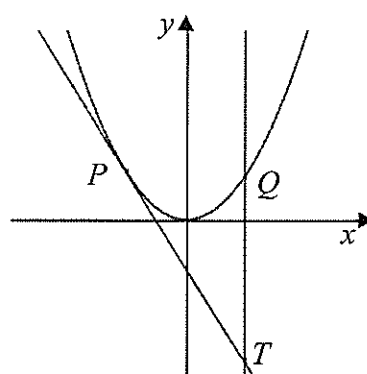
Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the volume of the solid of revolution formed when the region bounded by the curve $y = \frac{1}{\sqrt{4+x^2}}$, the x -axis, the y -axis and the equation $x = \frac{\pi}{2}$ is rotated about the x -axis. 3

- (b) Find the term independent of x in the expansion of $\left(3x^2 + \frac{2}{x}\right)^6$. 3

(c)



NOT TO
SCALE

Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.

The tangent at P and the line through Q parallel to the y axis intersect at point T .

The equation of the tangent at P is $y = px - ap^2$. (**Do NOT prove this**)

- (i) Find the coordinates of T . 1
- (ii) Write down the coordinates of M , the midpoint of PT . 1
- (iii) Determine the locus of M when $pq = -1$. 1
- (d) A sphere is being heated so that its surface area is expanding at a constant rate of 0.025 cm^2 per second. 3

Find the rate of change of the volume when the radius is 5 cm.

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Consider the function $f(x) = \frac{2x}{1-x^2}$.

(i) Show that the function is increasing for all values of x in its domain. 2

(ii) Sketch the graph of $y = f(x)$ showing the intercepts on the axes and any asymptotes. 3

(iii) Hence, or otherwise, find the values of k such that $\frac{2x}{1-x^2} = k$ has two solutions. 1

(b) Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos^2 2x}{x^2}$. 2

(c) Peter and his brother James are having a card night for themselves and 6 other friends. If they are to be seated at a round table, what is the probability that Peter and James do NOT sit next to each other? 2

(d) Find the maximum value of $2x(1-x)$ and hence determine the range of $y = \sin^{-1}[2x(1-x)]$ for $0 \leq x \leq 1$. 2

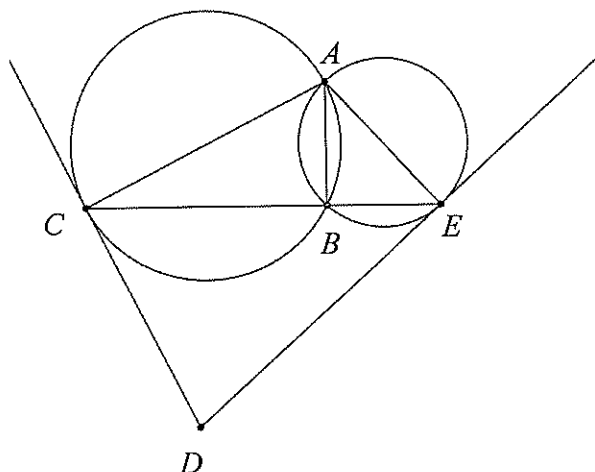
Question 5 (12 marks) Use a SEPERATE writing booklet.

Marks

- (a) Use mathematical induction to prove that $4^n > 2n + 1$, where n is a positive integer.

3

(b)



NOT TO
SCALE

In the diagram above, two circles intersect at A and B.
Points C and E lie on the circles and C, B and E are collinear.
Tangents at C and E meet at D.

3

Show that quadrilateral AEDC is concyclic.

- (c) The population, P , of a mining town after t years satisfies the equation

$$\frac{dP}{dt} = k(P - 1000).$$

The population was initially 10 000, and after five years it had decreased to 8 000.

- (i) Show that $P = 1000 + Ae^{kt}$ is a solution of the equation.

1

- (ii) Find the value of A .

1

- (iii) Find the value of k . (Give your answer correct to 3 significant figures)

1

- (iv) Find the number of years taken for the population to reach 5000.

2

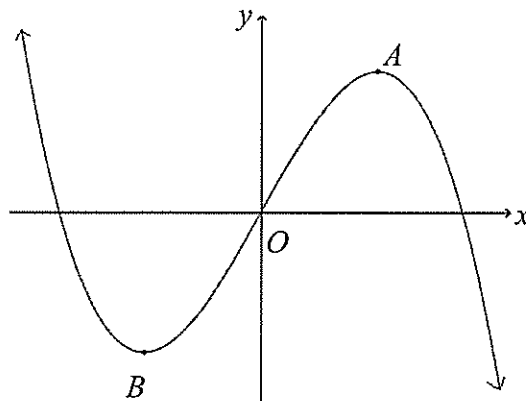
- (v) Sketch the graph of the population against time.

1

Question 6 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) The graph of $y = f(x)$, where $f(x) = 3x - x^3$, is shown in the diagram below.



NOT TO
SCALE

- (i) Find the coordinates of the turning points A and B . 2
- (ii) Find the largest domain containing the origin for which $f(x)$ has an inverse function. 1
- (iii) By considering the graph of $y = f(x)$, find the domain of $f^{-1}(x)$. 1
- (iv) By considering the gradient of $y = f(x)$, or otherwise, find the gradient of the inverse function $y = f^{-1}(x)$ at $x = 0$. 2
- (b) A particle moves in a straight line, so that its acceleration x cm from the origin is given by $\frac{d^2x}{dt^2} = 15 - 25x$. Initially the particle is at rest 1.6 cm to the right of the origin.
- (i) Show that the speed is given by $s = \sqrt{30x - 25x^2 + 16}$. 2
- (ii) Given that the motion is simple harmonic, find the interval in which the particle moves. 2
- (iii) Find the maximum speed, and the displacement where this occurs. 2

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Two of the roots of the equation $x^3 - ax^2 + b = 0$ are reciprocals.

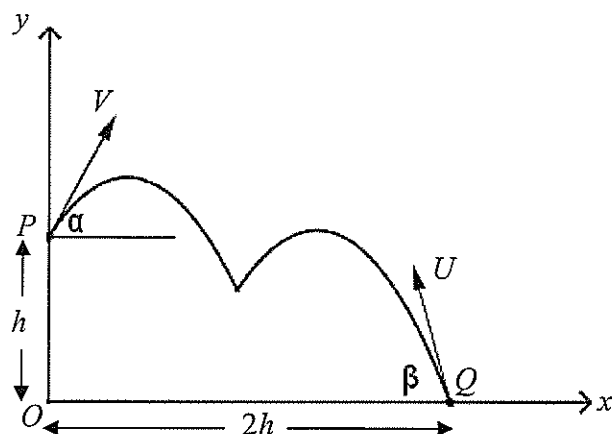
(i) Show that the third root is equal to $-b$. **2**

(ii) Show that $a = \frac{1}{b} - b$. **2**

(iii) Show that the 2 roots, which are reciprocals, will be real if $\frac{-1}{2} \leq b \leq \frac{1}{2}$. **2**

Question 7 continues on page 10

(b)



NOT TO
SCALE

O and Q are two points $2h$ metres apart on horizontal ground.

P is a point h metres directly above O . Particle A is projected from P towards Q with speed V m/s at an angle α above the horizontal.

At the same time particle B is projected from Q towards P with speed U m/s at an angle β above the horizontal. The two particles collide T seconds after projection.

For particle A the equations of motion are: $\ddot{x}_p = 0$ and $\ddot{y}_p = -g$.

It is known that its horizontal distance x_p from O , is given by:

$$x_p = Vt \cos \alpha, \text{ where } t \text{ is time in seconds. (Do NOT prove this)}$$

- (i) Use calculus to show that at time t seconds, its vertical distance y_p from O is given by: 2

$$y_p = Vt \sin \alpha - \frac{1}{2}gt^2 + h$$

- (ii) For particle B , write down expressions for its horizontal distance x_Q from Q and its vertical distance y_Q from Q at time t seconds. 2

- (iii) By considering the point of collision, find an expression for $\frac{V}{U}$ in terms of α and β . 2

End of paper

Question 1.

(a) $A(-3, 4)$ $B(1, -2)$ externally $\frac{m}{1} : \frac{n}{3}$

$$X = \frac{1x_1 - 1 - 3x_2 - 3}{1 - 3}$$

$$= \frac{1 + 9}{-2}$$

$$= -5$$

$$Y = \frac{1x_1 - 2 - 3x_2}{1 - 3}$$

$$= \frac{-14}{-2}$$

$$= 7$$

$$(-5, 7) //$$

(b) $\frac{d}{dx} \cos^{-1}(x^3) = \frac{-1}{\sqrt{1-x^6}} \times 3x^2$

$$= \frac{-3x^2}{\sqrt{1-x^6}} //$$

(c) $(x+1)$ is a factor if $P(-1) = 0$

$$P(-1) = (-1)^4 - (-1)^3 + k(-1) - 4$$

$$= 1 - (-1) - k - 4$$

$$= -2 - k$$

$$= 0 \text{ when } k = -2 //$$

$$\therefore P(x) = x^4 - x^3 - 2x - 4$$

if $P(x-2)$ is a factor $P(2) = 0$

$$P(2) = 2^4 - 2^3 - 2 \times 2 - 4$$

$$= 16 - 8 - 4 - 4$$

$$= 0 //$$

(d) $\frac{x+4}{x-2} \geq 3$

$$(x+4)(x-2) \geq 3(x-2)^2$$

$$(x-2)[(x+4) - 3(x-2)] \geq 0$$

$$(x-2)[4 - 2x] \geq 0$$

$$2(x-2)(5-x) \geq 0$$

$$2 < x \leq 5 //$$

(e) $\int_0^1 \frac{4x}{2x+1} dx$

$$= \int_1^3 \frac{2u-2}{u} \cdot \frac{du}{2}$$

$$= \int_1^3 \frac{u-1}{u} \cdot du$$

$$= \int_1^3 \left(1 - \frac{1}{u}\right) du$$

$$= \left[u - \ln u\right]_1^3$$

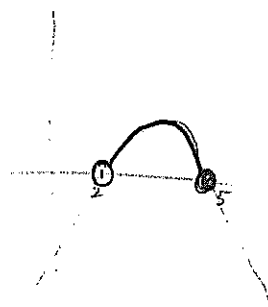
$$= (3 - \ln 3) - (1 - 0)$$

$$= 2 - \ln 3 //$$

$u = 2x+1 \Rightarrow 2x = u-1$
 $4x = 2u-2$
 $\frac{du}{dx} = 2 \Rightarrow dx = \frac{du}{2}$

$$x=1 \quad u=3$$

$$x=0 \quad u=1$$



Q2. a) $3\cos x + 5\sin x = 5 \quad 0^\circ \leq x \leq 360^\circ$

$$R = \sqrt{9+25} = \sqrt{34} \quad \tan \alpha = \frac{5}{3} \quad \text{ie } \alpha = 59^\circ 2'$$

$$\text{ie } \sqrt{34} \cos(x - \alpha) = 5$$

$$\cos(x - \alpha) = \frac{5}{\sqrt{34}}$$

$$x - 59^\circ 2' = 30^\circ 58', 329^\circ 2'$$

$$\therefore x = 90^\circ, 388^\circ$$

$$\text{since } 0 \leq x \leq 360^\circ, \quad x = 90^\circ, 388 - 360 = 28^\circ$$

$$\text{ie } x = 90^\circ, 28^\circ$$

b) $y = x^2$

$$y = 2x$$

$$\frac{dy}{dx} = 2x$$

$$m = 2$$

$$\text{at } x=2, m_1 = 4$$

$$\therefore m_2 = 2$$

$$\tan \theta = \left| \frac{4-2}{1+4 \times 2} \right|$$

$$= \frac{2}{9}$$

$$\therefore \theta = 12^\circ 32'$$

c) $\int 2\sin^2 x = \int 1 - \cos 2x \, dx$
 $= x - \frac{1}{2} \sin 2x + C$

d) i) $P(B) = 0.05 \quad P(\bar{B}) = 0.95$

$$P(1B) = {}^{10}C_1 (0.05)(0.95)^9$$

$$= 0.31512$$

ii) $P(\geq 2B) = 1 - [P(0B) + P(1B)]$

$$= 1 - [0.95^{10} + 0.31512]$$

$$= 0.0861$$

e) $x(x+5) = 6^2$

$$PS, PR = OP^2$$

$$x^2 + 5x - 36 = 0$$

$$(x+9)(x-4) = 0$$

$$x = -9, 4$$

$$\therefore x = 4 \quad (x \geq 0)$$

Question 3 $\frac{\pi}{2}$

(a) $V = \pi \int_0^{\frac{\pi}{2}} y^2 dx$

$$= \pi \int_0^{\frac{\pi}{2}} \frac{1}{4+x^2} dx$$

$$= \frac{\pi}{2} \left[\tan^{-1} \frac{x}{2} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} \left[\tan^{-1} \frac{\pi}{4} - \tan^{-1} 0 \right]$$

$$= \frac{\pi}{2} \tan^{-1} \frac{\pi}{4} \text{ (exact) } \textcircled{3}$$

(1.05 2dp)

$\frac{dv}{dt} = ?$ $\frac{ds}{dt} = 0.025$ at $r=5$

(d) $\frac{dv}{dt} = \frac{ds}{dt} \times \frac{dv}{ds}$ } Need $\frac{dv}{ds}$

$$= 0.025 \times \frac{r}{2}$$

$$= 0.025 \times \frac{5}{2}$$

$$= 0.0625 \text{ cm}^3/\text{s}$$

$\frac{dv}{ds} = \frac{dv}{dr} \times \frac{dr}{ds}$

$$= 4\pi r^2 \times \frac{1}{8\pi}$$

$$= \frac{r}{2}$$

$$\boxed{S = 4\pi r^2}$$

$$\boxed{\frac{ds}{dr} = 8\pi r}$$

$$\boxed{V = \frac{4}{3}\pi r^3}$$

$$\boxed{\frac{dv}{dr} = 4\pi r^2}$$

(b) $\left(3x^2 + \frac{2}{x}\right)^6$ has general term $\binom{6}{r} (3x^2)^{6-r} \left(\frac{2}{x}\right)^r$

$$= \binom{6}{r} 3^{6-r} x^{12-2r} \cdot 2^r \cdot x^{-r}$$

$$= \binom{6}{r} 3^{6-r} \cdot 2^r \cdot x^{12-3r}$$

term independent of x (power of $x=0$)

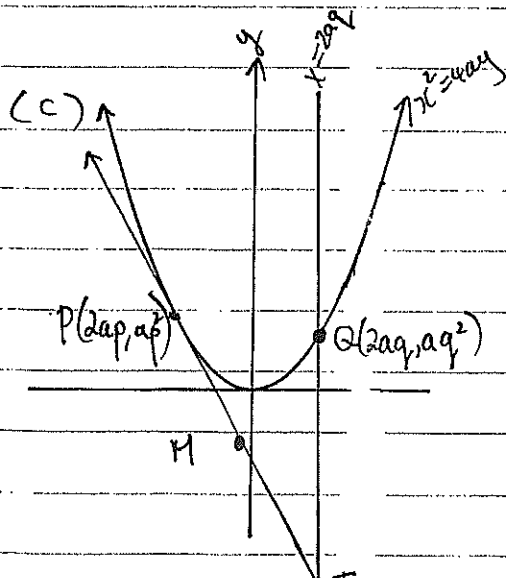
$$\therefore 12-3r=0$$

$$3r=12$$

$$r=4$$

$$\therefore \binom{6}{4} 3^{6-4} \cdot 2^4 \cdot x^0 = 15 \times 9 \times 16$$

$$= 2160 \text{ // } \textcircled{3}$$



(i) Finding T: $x=2aq$ satisfies $y=px-ap^2$

$$\therefore y=2apq-ap^2$$

$$T(2aq, 2apq-ap^2) \text{ //}$$

(ii) $M_{PT} = \left(\frac{2ap+2aq}{2}, \frac{ap^2+(2apq-ap^2)}{2} \right)$

$$= (a(p+q), apq) \text{ //}$$

(iii) given $pq=-1$ then $y=apq$

$$= -a$$

$\therefore M$ lies on the directrix of $x^2=4aq$

Question 4 Solutions.

$$f(x) = \frac{2x}{1-x^2} \quad x \neq \pm 1$$

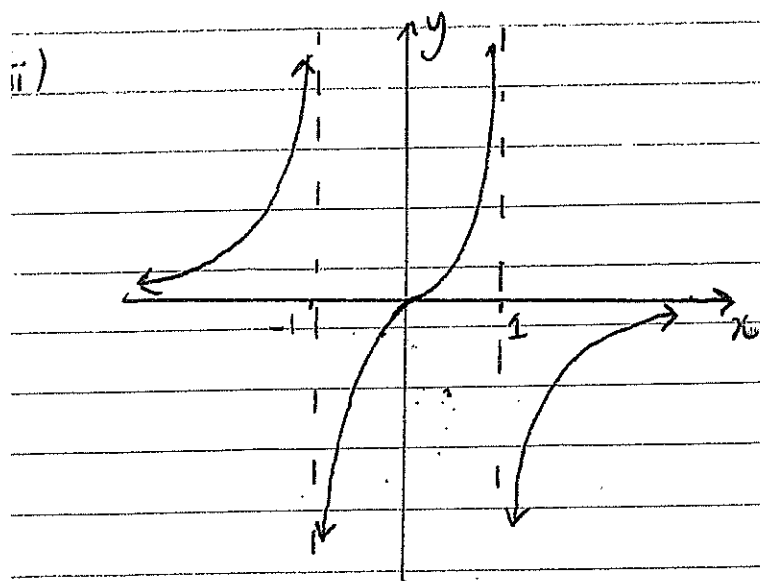
$$f'(x) = \frac{2(1-x^2) + 4x^2}{(1-x^2)^2}$$

$$= \frac{2x^2 + 2}{(1-x^2)^2}$$

$$2x^2 + 2 > 0 \text{ for all } x$$

$$(1-x^2)^2 > 0 \text{ for all } x, x \neq \pm 1$$

$\therefore f'(x) > 0$ for all x in domain
 $f(x)$ is increasing.



ii) From graph, all $k, k \neq 0$.

$$\begin{aligned} \text{b) } \lim_{x \rightarrow 0} \frac{1 - \cos^2 2x}{x^2} &= \lim_{x \rightarrow 0} \frac{\sin^2 2x}{x^2} \\ &= 4 \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right)^2 \\ &= 4 \end{aligned}$$

$$\begin{aligned} \text{c) Arrangements not together} &= 1 \times 5 \times 6! \\ \text{Total arrangements} &= (8-1)! \\ &= 7! \end{aligned}$$

$$\begin{aligned} \therefore P(\text{not together}) &= \frac{1 \times 5 \times 6!}{7!} \\ &= \frac{3600}{5040} \\ &= \frac{5}{7} \end{aligned}$$

$$\text{d) For } 0 \leq x \leq 1,$$

$$y = 2x(1-x) = 2x - 2x^2$$

$$\frac{dy}{dx} = 2 - 4x$$

$$\text{let } \frac{dy}{dx} = 0$$

$$x = \frac{1}{2}$$

$$\text{when } x = \frac{1}{2}, \text{ max value is } \frac{1}{2}$$

$$\begin{aligned} \therefore \text{Range of } f(x) &= \sin^{-1}(2x(1-x)) \\ \text{is } 0 &\leq f(x) \leq \frac{\pi}{6} \end{aligned}$$

QUESTION 5

a) Prove true for $n=1$

$$L.H.S. = 4$$

$$R.H.S. = 2+1$$

$$= 3$$

$$\therefore L.H.S. > R.H.S.$$

$\therefore$ True for $n=1$

Assume true for $n=k$

$$\text{i.e. } 4^k > 2k+1$$

Prove true for $n=k+1$

$$\text{i.e. } 4^{k+1} > 2(k+1)+1$$

$$4^{k+1} > 2k+3$$

$$\text{now } 4^k > 2k+1$$

$$\text{H. } 4^k > 4(2k+1)$$

$$4^{k+1} > 8k+4$$

$$4^{k+1} > 2k+3 + 6k+1$$

$$\therefore 4^{k+1} > 2k+3$$

$\therefore$ True for $n=k+1$

Since true for $n=1$ etc.

$$A = 9000$$

iii) when $t=5$ $P = 8000$

$$8000 = 1000 + 9000 e^{5k}$$

$$e^{5k} = \frac{7}{9}$$

$$5k = \ln\left(\frac{7}{9}\right)$$

$$k = -0.0503$$

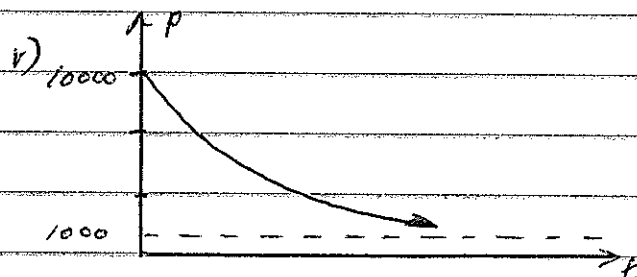
iv) when $P = 5000$

$$5000 = 1000 + 9000 e^{kt}$$

$$e^{kt} = \frac{4}{9}$$

$$kt = \ln\left(\frac{4}{9}\right)$$

$$t = 16.12 \text{ years}$$



b) $\angle DCE = \angle CAB = x$ ($\angle$ in alternate segment)

$\angle DEC = \angle EAB = y$ ($\angle$ in alternate segment)

$$\therefore \angle CAE = x+y$$

$$\angle CDE + x + y = 180 \text{ (angle sum } \triangle CED)$$

$$\therefore \angle CDE + \angle CAE = 180$$

$\therefore$ AEDC cyclic quadrilateral (opposite $\angle$'s supplementary)

c) i) $P = 1000 + Ae^{kt} \Rightarrow Ae^{kt} = P - 1000$ — (1)

$$\frac{dP}{dt} = kAe^{kt}$$

$$= k(P-1000) \text{ by substituting (1)}$$

ii) when $t=0$, $P = 10000$

$$10000 = 1000 + Ae^0$$

Question 6

$$f(x) = 3x - x^3$$

$$f'(x) = 3 - 3x^2$$

$$\text{let } f'(x) = 0$$

$$0 = 3 - 3x^2$$

$$x^2 = 1$$

$$x = \pm 1$$

$$\text{when } x = 1, f(1) = 3 - 1 = 2$$

$$x = -1, f(-1) = -3 - (-1)^3 = -2$$

$$\therefore A = (1, 2); B = (-1, -2)$$

i) longest domain: $-1 \leq x \leq 1$ containing $(0, 0)$

iii) Range of $f(x)$ for restricted domain is $-2 \leq y \leq 2$

$\therefore$ Domain of $f^{-1}(x)$ is $-2 \leq x \leq 2$

$$iv) f'(x) = 3 - 3x^2$$

$$f'(0) = 3$$

$$\therefore \frac{dx}{dy} = \frac{1}{3}$$

Gradient of inverse function at $x=0$ is $\frac{1}{3}$.

$$(b) \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 15 - 2x$$

$$\frac{1}{2} v^2 = \int (15 - 2x) dx$$

$$(b) \frac{1}{2} v^2 = 15x - \frac{25x^2}{2} + C$$

$$\text{when } t=0, x=1.6, v=0$$

$$\therefore 0 = 15(1.6) - \frac{25}{2}(1.6)^2 + C$$

$$C = 8$$

$$\cancel{v^2 = 30x - 25x^2 + 16}$$

$$v^2 = 30x - 25x^2 + 18$$

$$s = \sqrt{30x - 25x^2 + 18}$$

$$(ii) \text{ let } s = 0$$

$$0 = 30x - 25x^2 + 18$$

$$x = \frac{20 \pm \sqrt{900 + 4(25)(16)}}{50}$$

$$= \frac{30 \pm 50}{50}$$

$$x = -\frac{2}{5}, \frac{8}{5}$$

Particle moves on interval

$$-\frac{2}{5} \leq x \leq \frac{8}{5}$$

$$(iii) \text{ Max speed at } x = \frac{-\frac{2}{5} + \frac{8}{5}}{2}$$

$$= \frac{3}{5}$$

$$s = \sqrt{30\left(\frac{3}{5}\right) - 25\left(\frac{3}{5}\right)^2 + 18}$$

$$= \sqrt{25}$$

$\therefore$ Max speed is 5 cm s^{-1} when $x = \frac{3}{5} \text{ cm}$ to right of origin.

Q7a) $x^3 - ax^2 + b = 0$

i) let roots be $x, \frac{1}{x}, \beta$

Product of roots: $x \cdot \frac{1}{x} \cdot \beta = -\frac{d}{a}$

$\therefore \beta = -b$

ii) $-b$ is a root

$\therefore P(-b) = (-b)^3 - a(-b)^2 + b = 0$

$-b^3 - ab^2 + b = 0$

$ab^2 = -b^3 + b$

$a = -b + \frac{1}{b}$

iii) $x^3 - \left(b - \frac{1}{b}\right)x^2 + b = 0$

Sum of roots 2 @ a time: $x \cdot \frac{1}{x} + x \cdot (-b) + \frac{(-b)}{x} = \frac{c}{a} = 0$

$\therefore 1 - xb - \frac{b}{x} = 0$

$x - x^2b - b = 0$

ie $bx^2 - x + b = 0$

real roots $\Delta \geq 0$

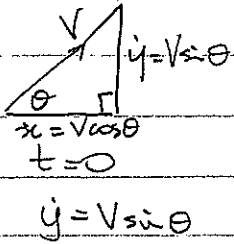
$1 - 4b^2 \geq 0$

$(1 - 2b)(1 + 2b) \geq 0$

ie $-\frac{1}{2} \leq b \leq \frac{1}{2}$

Q7b) i)

$$\ddot{y}_p = -g$$



$$\dot{y}_p = \int -g \, dt$$

$$= -gt + C_1$$

$$\dot{y}_p = -gt + V \sin \alpha$$

$$y = \int -gt + V \sin \alpha \, dt$$

$$y = -\frac{gt^2}{2} + Vt \sin \alpha + C_2$$

$$t=0$$

$$y = h$$

$$\therefore y = -\frac{gt^2}{2} + Vt \sin \alpha + h$$

ii)

$$x_a = 2h - ut \cos \beta$$

$$y_a = -\frac{gt^2}{2} + ut \sin \beta$$

iii)

$$2h - ut \cos \beta = Vt \cos \alpha$$

$$Vt \cos \alpha + ut \cos \beta = 2h$$

$$t = \frac{2h}{V \cos \alpha + u \cos \beta}$$

$$V \cos \alpha + u \cos \beta$$

$$-\frac{gt^2}{2} + Vt \sin \alpha + h = -\frac{gt^2}{2} + ut \sin \beta$$

$$h = ut \sin \beta - Vt \sin \alpha$$

$$t = \frac{h}{u \sin \beta - V \sin \alpha}$$

$$u \sin \beta - V \sin \alpha$$

$$\frac{2h}{V \cos \alpha + u \cos \beta} = \frac{h}{u \sin \beta - V \sin \alpha}$$

$$V \cos \alpha + u \cos \beta$$

$$u \sin \beta - V \sin \alpha$$

$$2u \sin \beta - 2V \sin \alpha = V \cos \alpha + u \cos \beta$$

$$V \cos \alpha + 2V \sin \alpha = 2u \sin \beta - u \cos \beta$$

$$V(\cos \alpha + 2 \sin \alpha) = u(2 \sin \beta - \cos \beta)$$

$$\frac{V}{u} = \frac{2 \sin \beta - \cos \beta}{\cos \alpha + 2 \sin \alpha}$$